IKT200-Assignment 2- Solution

Task 1: Agents

Explain in your own words what is meant by an agent (in the field of machine learning).

An agent is an autonomous entity which observes (through sensors) and trades (through actuators) against an environment for achieving its goals. Intelligent agents can also learn or use knowledge to achieve their goals.

Agents can be very simple or very complex. Russell and Norvig defined it as " … anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators".

A human agent has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators. A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators. A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on screen, writing files, and sending network packets.

Task 2: Finite State Machines

We have learned about Finite State Machines, and various ways in which to make transitions.

a) Explain in their own words how transitions in active state generates and child-states compete models differ.

Active state generates: Here it is the active state that determines transitions to the other state. If the vacuum cleaner has found dust, it first checks if it has enough power. If it does not have it, it exposes it to vacuum up and goes to recharge, if it has enough power it starts to ride. Here, state 2 decides whether to return to state 1 or state 3.

Child-states compete: Here, a relevance system is used. All nodes are asked how relevant it is for them to become active state. Whoever answers with the highest score "wins".

b) Write pseudocode (on paper) or java code (on paper or in suitable tool) for a finite state machine (minimum 2 states, preferably several).

```
main ()
 {
    // Set default value
    currentState = 1;
    while (true) {
            switch currentState {
            case 1:
                    // Check to make a transition:
                    if (enterState2 == true) {
                            stateSwitch (2) ;
                            break;
                    }
                     // *** Make state 1 specific content here. ***
                     break;
             case 2:
```

```
                        // Check to make a transition:
                        if (enterState1 == true) {
                                stateSwitch (1);
                                break;
                        }
                         // *** Make state 2 specific content here. ***
                         break;
                }
        }
}
```

Problem 3: IoT design

Answer the following questions (Use textbook on Chapter 2)

a) Industrial design is much more than the appearance of an object. What is included in industrial design? Mention as many items as you come across.

   Industrial design is a creative activity that aims to develop and enhance the quality of series-made objects, processes, services and systems. The strength of the industrial designer is the ability to think innovatively and holistically- user interface selection, making the object easy to use, functionality, service design, determine how the object is constructed, etc.

b) What is meant by quiet technology? ("Calm technology")

   Quiet technology is a term used by Mark Weiser when he mentions the forerunner to the Internet of Things: Ubiquitous computing ("ubicomp"). By quiet technology is meant things with processor power available, but which do not attract attention when not actively used. Power and networks are a challenge for such devices. Another challenge is configuration and user interaction.

c) What must we take care of when we build solutions that exchange personal data?

   Safety! Remember, for example data encryption. (One-way encryption, preferably with salt where possible.)

   Privacy! Who owns the photos collected by the security company surveillance camera hanging on your house wall and filming the rest of the street? Who owns photos captured by a camera in public space?

d) What is Postel's Law about?

   Postel's Law is about being conservative/accurate in what you do yourself and being liberal/receptive in what you accept from others.

Problem 4: Design Principles

Explain the following design principles in your own words:

a) Loose/low coupling.

   Ensure that the components are dependent of each other to the minimum extent. Few possible couplings required.

b) Use existing standards.

Look for others to use the solution, Easier to find work force that can contribute to further development, Easier to reuse components, Greater chance of the system staying compatible with other relevant solutions.

c) Use formats that have a tradition of staying backwards compatible.

   If you use backward compatible formats, it is easier to get more things to talk together. Not all formats are updated.

d) Graceful degradation.

   Keep most of the system up although some elements stop working. If your refrigerator has an LCD screen, then the refrigerator must still be able to function even if the LCD screen is damaged.

e) Preservation of functionality.

   When something is upgraded, strive to safeguard the user experience. For example, stereo system or dimmer switch with remote volume/dimmer button.

f) Recognize value.

   Use known components for known tasks. Big round button on the stereo to control the volume and not allow the mini jack to supply power.